Cloud computing

1. Cloud computing is the delivery of computing services over the internet, or "the cloud". It allows users to access computing resources like storage, servers, and software on-demand.

2. Scalability, security, Accessibility, security, disaster recovery

3. AWS, digital ocean, GCP

4. security, privacy, cost management

5. internet based, pay as you go model, managed by cloud provider, access from anywhere

Cloud Models: IaaS, PaaS, SaaS

1. Infrastructure as a Service — virtual servers, storage.
Use case: Hosting websites, running apps.

2.Heroku, Vercel

3. SaaS vs. PaaS:

- **SaaS:** Ready-to-use software (e.g., Gmail).
- **PaaS:** Platform for app development (e.g., Azure App Service).

4. SaaS

5. Real-world examples:

- **IaaS:** Hosting VMs in Azure
- **PaaS:** Deploying apps via Azure App Service
- **SaaS:** Using Microsoft 365

Public, Private, and Hybrid Cloud

1. A public cloud is a type of cloud computing where resources, like computing power, storage, and software, are provided by a third-party over the internet and are shared among multiple organizations and individuals. Managed by cloud providers.

2. Private cloud: Dedicated to one organization, managed internally or by a provider.

3. Hybrid cloud: Combines public and private clouds.

- **Advantage:** Flexibility
- **Disadvantage:** Complex management

4. Best for sensitive data: Private cloud

5. Multi-cloud vs. Hybrid cloud:

- **Multi-cloud:** Uses multiple public clouds (e.g., Azure + AWS).
- **Hybrid cloud:** Mix of public + private clouds.

Microsoft Azure

1. A cloud platform offering services for computing, storage, AI, and more.

2. Azure vs. AWS:

- **Azure:** Strong integration with Microsoft products.
- **AWS:** Broader service range.

3. Key Azure services:

- Virtual Machines (VMs)
- Blob storage
- Azure SQL Database

4. AI/ML in Azure: Yes, e.g., **Azure Machine Learning** for training AI models.

5. Benefits of Azure:

- Scalability
- Security
- Global reach
- Integration with Microsoft tools

Azure Global Infrastructure

1. Azure Region: A set of data centers in a specific location (e.g., East US).

2. Number of regions: ~60+ globally.

3. Availability Zone: Physically separate data centres within a region to improve fault tolerance.

4. How they improve reliability: Ensure apps stay online during failures by distributing resources.

5. Region Pair vs. Availability Zone:

- **Region Pair:** Two geographically distant regions for disaster recovery.
- **Availability Zone:** Multiple data centres within one region for high availability